02 | Use of the regular-coach interior

Layout	Cabin outside / inside width	Seats	Clear floor	End zone*
A - narrow longitudinal	24 ft / 90 in	24	176.3 ft²	21 in
B - shorter transverse	24 ft / 104 in	28	203.7 ft²	35 in
C - preferred mockup	26 ft / 104 in	32	221.0 ft²	31 in
D - higher-capacity test	28 ft / 104 in	36	238.3 ft²	27 in

*End zones are seat-to-wall dimensions, not guaranteed unoccupied vestibules. All transverse rows face the same direction. With the illustrative knee envelope used here, C leaves about 21.5 in beyond the last-row knees, and D about 17.5 in. Feet and posture can reduce these further. The longitudinal center aisle remains geometrically continuous.

Preferred C: explicit dimensional chain

Exterior cabin length 312 in - two provisional 3 in end walls = 306 in clear length. Eight rows at 32 in pitch, with a 20 in total seat/back envelope, use 7 × 32 + 20 = 244 in. The remaining 62 in provides 31 in at each end. Across the car: 40 in bench + 24 in aisle + 40 in bench = 104 in. Each double bench assigns 20 in per adult; no armrests or tables.

Thirty feet of bench length has two meanings. C provides 8 × 40 in = 26 ft 8 in of cumulative bench width on each side. D provides 9 × 40 in = exactly 30 ft per side. A continuous 30 ft longitudinal bench would require a different, longer cabin. D is retained for comparison; its greater count has not yet earned the longer/heavier car.

Comfort and evacuation must set the actual capacity

A NASA-hosted anthropometry table provides useful study comparators: 18.3 in seated hip breadth, 22.1 in shoulder breadth and 27.5 in buttock-to-knee length at the upper values shown [S07]. It is not a general-public rail accommodation standard. Twenty inches per person does not guarantee winter-coat comfort. At 32 in pitch, a 2 in illustrative back/buttock offset leaves only 2.5 in static knee clearance against that comparator.

Build two adjacent full-size rows plus the last row and both door approaches. Use six varied adults in winter clothing; the drawing set shows four seated and two standing for scale. Test passing, assistance, standing up, luggage, door hardware and evacuation in both directions. Standing figures do not authorize standing passengers while moving. Reduce seats or revise width/pitch if the trial fails.

03 | Accessible coach, boarding and end platforms

The preferred accessible coach uses the same 26 ft cabin and 104 in clear width. It has **11 fixed seats plus two wheelchair positions**. The three-seat reduction from the comparison 16-place layout preserves a continuous 36 in interior route to both ends. This is a packaging choice pending the authority-agreed accessibility basis, not a ruling that the alternative is unlawful.

Space reservation	Coordinates and purpose, inches
Coordinate system	x = front inside wall toward rear, 0 to 306; y = centerline, -52 to +52. Finished floor is z = 0.
Front turning/boarding	x 0-60 across the cabin; 60 in circle centered at (30, 6). Opposed 42 in clear side openings, x 9-51.
Continuous route	x 60-306, y -12 to +24; 36 in end openings on the same 6 in offset center.
Wheelchair reservations	Two 36 × 60 in bays: x 60-120 and 120-180, y -52 to -16. Four-inch buffer between bays and route.
Companions	Two 24 in singles opposite the wheelchair bays; no fold-down seat counted within an occupied wheelchair place.
Rear seating	Three rows of one 40 in double plus one 28 in single; 32 in pitch. Rear seat-to-wall zone 32 in, partly occupied by legs.

Complete boarding and rescue are unresolved release gates

Prefer a controlled level or bridged station interface after vehicle floor height, suspension travel and platform gap are established. Compare a fixed-platform bridge with a rated lift if the site requires it. The 28 in floor-above-rail value in the packaging model is provisional; it does not set a compliant platform height. Side doors, headers, lift mass and manual operation must be coordinated before frame release.

At a disabled train, provide a practical route from each occupied wheelchair bay to a designated rescue interface, rated boarding/evacuation equipment, enough trained attendants, and access from the route to a safe area. The rear 36 in interior route alone does not create an accessible way off the end platform. Stairs alone are not a wheelchair evacuation solution. Securement and occupant restraint decisions follow the actual vehicle classification and equipment review [S06].

Both regular and accessible cars retain onboard end-platform/stair intent. Protect moving couplings below with engineered barriers and maintenance access; gates and doors must not sweep into reserved passenger space. Keep stair stowage, ground clearance, guards, slip resistance, lighting and traction interlocks in the next detail package. No tread, riser, pin or guard thickness is released here.

05 | Mass, axle loads and the 36-inch gauge decision

Mass allocation	Regular, kg	Accessible, kg
Underframe / draft structure	1,500	1,550
Two bogies / wheelsets / suspension	1,800	1,800
Brakes / controls / piping	250	250
Body / roof / glazing / doors	900	950
Floor / insulation / finish	400	420
Seats / fixtures	180	100
Platforms / stairs / guards	400	430
HVAC / ventilation / ducts	550	450
Electrical / batteries / lighting	240	280
Allocated intercar hardware	180	180
Boarding / securement / rescue	-	440
Subtotal + 10% growth	6,400 + 640	6,850 + 685
Tare allocation	7,040	7,535

These are subsystem budget allocations, not verified component weights or structural takeoffs. Three regular cars + one accessible car + 14,500 kg locomotive/hotel allowance = **43,155 kg tare**. Add 113 people at 100 kg and 300 kg for two mobility devices: **54,755 kg**. At 145 kg/person: **59,840 kg**. Wheelchair users count once among people; devices are additional. The 145 kg value is a stress input, not a guest weight limit.

The four crew are conservatively placed one per coach for local mass, drawbar and HVAC screens. They are four total, not four attendants plus an uncounted driver. Final staffing must include the driver and actual crew locations. Confirm fuel/fluids within the locomotive allowance and actual device/equipment masses.

A broader tare range gives a loaded stress case up to 63,685 kg; retain 55/60/65 t performance screens. The nominal regular car is 10,340 kg, stress 11,825 kg. On four equally loaded axles, the latter averages 2.96 t/axle; this is not a design axle rating. Uneven people, equipment, suspension and dynamics can increase individual loads.

The 104 in cabin is 15.6% wider than the earlier 90 in concept. Gauge remains 36 in only as a candidate. A vehicle engineer must establish actual CG, wheel unloading, sway/roll, crosslevel and wind effects, full/partial occupancy, curves and buff forces. Bought-in engineered bogies are preferred for quotation. Wheel profile, back-to-back, flange, rail-head and turnout/check-rail geometry form one interface; matching nominal gauge is insufficient.

06 | Traction, braking and connection loads

At 2% uphill grade, resistance coefficient 0.005 and 0.10 m/s² acceleration, required wheel force is M[g(0.02 + 0.005) + 0.10]. At 7.5 mph and assumed 75% shaft-to-wheel efficiency, nominal/stress cases need 18.90/20.65 kN and 84.5/92.3 kW shaft power. Add local auxiliaries, curve/start resistance and engine derating; hotel power is separate. These are demand screens, not a selected engine or locomotive rating.

At assumed adhesion 0.15, required driven mass is about 12.85/14.04 t. At 0.10, it is 19.27/21.06 t. A 14.5 t locomotive assembly is therefore marginal in the 0.15 stress case even if all its weight is driven, and fails the 0.10 screen. More engine power does not fix insufficient adhesion.

Brake performance targets to develop with a brake supplier

Train mass	Force for 0.5 m/s² net decel on 2% descent	Energy in screen	3% gravity hold only
55 t	38.29 kN	0.537 MJ	16.18 kN
60 t	41.77 kN	0.586 MJ	17.65 kN
65 t	45.25 kN	0.635 MJ	19.12 kN

All three masses give 21.12 m (69.3 ft) in the same ideal model: start at 7.5 mph, accelerate freely downhill during a total 2 s zero-brake delay, then achieve 0.5 m/s² net deceleration. Speed reaches 8.38 mph before braking. Add human reaction if the 2 s input represents hardware latency alone. Energy includes application speed and downhill travel during braking; rotating inertia, gradual force build-up and design margins are excluded.

The target requires at least 0.071 effective adhesion with all weight braked, before margins or imbalance. At assumed adhesion 0.05, the ideal stop grows to 101.5 ft; at 0.03, to 257.9 ft. These are sensitivities, not guaranteed wet-rail values. Holding forces above are bare gravity loads, not approved parking ratings.

Keep hydraulic service actuation in the comparison. An ordinary pressure-applied hydraulic circuit is not an adequate complete answer to loss of pressure, separation or parking. Compare independently stored-energy emergency/parking applications with supervised continuity, and a railway air-brake architecture with automatic separation response. Choose only after demonstrated service, emergency, parking and breakaway functions on all cars; do not change the owner preference without the comparison.

The first connection carries all trailing cars: 40.255 t nominal or 45.340 t stress. Illustrative stress demand is 20.10 kN tension on a 3% uphill start, and 36.01 kN compression if trailing brakes are ineffective while the leading system imposes 0.5 m/s² net deceleration on a 3% descent. That mismatch is a design load case, not a claimed stopping capability. Slack impact, fatigue, articulation, combined loads and retention can govern. No final bar, pin, clevis or weld size is assigned.

07 | HVAC, electrical power and equipment space

The enlarged cabin is 7.7724 × 2.6416 × 2.2 m clear, with 9 m² assumed glazing. Summer screen: 110°F outside / 75°F inside; winter: 30°F / 68°F. Study U-values are 0.7 W/m²K opaque and 2.7 glass; glass SHGC 0.5, incident solar 500 W/m² and roof sol-air increment 20 K. These are assumptions, not a verified site design-weather or code basis.

Thermal/electrical screen	Regular: 32 guests + 1 crew	Accessible: 13 guests + 1 crew
Cooling, dry outside-air case	16.96 kW thermal	10.49 kW thermal
Cooling, humid stress case	22.70 kW thermal	13.28 kW thermal
Winter loss before occupants/warmup	8.74 kW thermal	5.10 kW thermal
Heating capacity study range	9-12 kW thermal	6-9 kW thermal

Ventilation is assumed at 7.5 L/s/person, plus 0.03 m³/s time-averaged door/leakage air per car. People contribute 120 W each; lighting/electronics add 200 W/car. Humid stress adds 0.006 kg/kg humidity ratio to both fresh air and door/leakage, using 2,501 kJ/kg latent heat. A 15% thermal allowance is applied after summation. Formulas and contributions are in the workbook/source notes.

Train cooling is 61.35 kW dry or 81.39 kW humid thermal. At assumed delivered COP 2.5 plus 3 kW other load, electricity is 27.54/35.56 kW; at COP 2.0, humid demand becomes 43.69 kW. Carry 40-70 kW delivered-continuous hotel generation for study. Starting currents, derating, winter backup heat and protection remain unresolved; a 40 kW choice may fail the duty.

Real equipment reference	What it establishes / what is open
NW Rail NW-64-CN [S08]	Nominal five-ton condenser family (~17.6 kW thermal). Near the dry regular-car screen, below humid stress; delivered 110°F performance and matched ventilation are unverified.
Stadco Mark 6 [S09-S10]	42 model publishes 38 kW continuous; 80 model 70 kW continuous at reference conditions. Ambient/altitude derating and transient loads must be checked.
Generator packaging	Published packages are 30-32 in high. They cannot fit wholly below the provisional 28 in coach floor with structure and ground clearance. Study a segregated locomotive equipment bay.

A separate generator tender would add another rolling vehicle and require revised mass, length, brakes and couplings. Reserve protected emergency batteries independent of hotel-power failure. Coordinate flexible jumpers, grounding/bonding, protection, cable strain relief, exhaust/intake separation, condensate, noise and service access before envelope freeze.

11 | Curve envelopes and coupling coordination

Reserve 50 m radius as a planning reference while evaluating the identified Severn Lamb Jupiter. Its published catalog limits are 914 mm gauge, 50 m radius, 1% grade and 16 km/h maximum speed [S11]. Those figures do not rate this custom consist. The 2-3% calculations in this report are stress studies; they cannot be assigned to that catalog locomotive.

Radius	Body relative yaw	Per-pivot yaw	Inside platform corner gap	Outer reach from track CL
50.0 m	12.49°	6.25°	18.20 in	61.91 in
40.0 m	15.60°	7.80°	15.33 in	63.55 in
30.0 m	20.76°	10.38°	10.64 in	66.19 in
24.4 m	25.47°	12.74°	6.44 in	68.53 in

Calculated for equal 400 in-long, 110 in-wide rectangular coaches with 216 in bogie centers, 14 in pivot setbacks and a 58 in pin-center connection. Both bogie centers lie on the same circle; no roll, lateral suspension travel, slack, vertical transitions, stair projections, door movement or reverse curves are included. Positive gap does not establish a safe pinch clearance, passable gangway or approved curve radius.

Traceable geometry

With consistent units: $\rho = \sqrt{(R^2 - (B/2)^2)}$; $h = L/2 - s$; $\alpha = \operatorname{atan}(h/\rho) + \operatorname{asin}[d / (2\sqrt{(\rho^2+h^2)})]$. Relative body yaw = 2α; inside-end gap = $d - 2s\cos\alpha - W\sin\alpha$. Outer reach = $\sqrt{[(\rho+W/2)^2 + (L/2)^2]} - R$. This reach already includes half body width; do not add it again.

At 50 m, the inner reach is 57.96 in and outer reach 61.91 in before dynamic/design clearance. Station, scene, fence and crossing offsets must be larger by engineer-established allowances. The locomotive-to-first-coach geometry needs a separate model; equal-coach results cannot substitute for it.

Interfaces to close before drawings become fabrication details

Freeze common rail, floor, bolster and coupler datums; choose an engineered coupling system with documented tension/compression, fatigue, articulation, retention, wear and inspection criteria; then design its load path into both frames. Route brake hoses and power/control jumpers through the full swept envelope. Test with a physical articulation fixture before relying on concealed under-platform packaging.

The four study coach frames plus three study gaps occupy 140 ft 10 in before the locomotive and its connection. Station length, siding length and rescue access must be developed from the eventual complete train, with stopping tolerance and passenger circulation space added.

14 | Inspection, trials and release records

Hold point	Record / check	Release owner
HP01 - Basis	Authority/classification matrix; project responsibility chart; standards editions; permitted farm-crew scope.	Principal NV design professional / authorities
HP02 - Interior mockup	Six adults, winter clothing, last-row knees, both exits; representative wheelchairs, companions, doors, lift/bridge and rescue trial.	Vehicle/accessibility designers + operations
HP03 - Component interfaces	Manufacturer ratings, mass/CG, drawings, wheel/rail geometry, coupling travel and brake/electrical failure analysis.	Vehicle / track / brake integrators
HP04 - Pilot track	Material traceability, formation/compaction/drainage, rail/tie/fastener methods and measured productivity.	Track/civil engineer + required QA
HP05 - Hidden work	Survey levels, subgrade, drains, foundations, concealed welds/cables and inspections before covering.	Relevant engineer / inspector
HP06 - First vehicle	Material certificates, weld procedures/qualifications, inspection results, dimensions, mass and axle loads.	Structural/vehicle engineer + specialist inspector
HP07 - Static systems	Service/emergency/parking/breakaway functions, stowed boarding interlocks, electrical protection, emergency lighting and manual recovery.	System specialists / qualified inspector
HP08 - Dynamic / loaded	Low-speed envelope and clearance trials; empty/full/uneven loading, stopping, grade holding, temperature and fault tests under approved plan.	Vehicle/track/brake engineers
HP09 - Public release	As-built files, operating limits, maintenance and inspection manuals, staff competence, rescue drills and required certificates.	Owner + authorities / qualified inspector

Test speeds, payloads, proof loads, pass/fail tolerances and personnel protection must be set by the responsible professionals. The calculation targets here do not authorize improvised public trials or substitute for those criteria.

Crew daily record template

Date / weather / crew and equipment; chainage and drawing revision; materials and batch IDs; completed quantities and hours; gauge/alignment/crosslevel/twist checks against issued limits; rail-joint/fastener checks; inspections called and accepted; photographs before concealment; nonconformance number, correction and verifier; next work-front release. Keep construction QC separate from independent QA.

After the first pilot, update actual productivity, rework, equipment utilization and unit costs in the workbook. Before repetition, update the prototype vehicle weight ledger with measured components, not CAD volume estimates. These two feedback loops are the main controls on owner-built cost and lightweight construction.